

Hand vascular geometry analysis

Hand vascular geometry

- Hand vascular geometry analysis involves examining the patterns and structures of blood vessels in the hand
- typically using imaging techniques like infrared or near-infrared (NIR) imaging.
- can be applied in various fields, including biometrics for identity verification, medical diagnostics for detecting vascular diseases, and monitoring of blood flow in clinical settings. The main steps are as follows.

1) Imaging Techniques:

- **Infrared Imaging:** Hand veins are captured using infrared light, which penetrates the skin and highlights the subcutaneous blood vessels. This is commonly used in biometric systems.
- **Near-Infrared Spectroscopy (NIRS):** NIRS can be used for more detailed analysis of blood oxygenation and hemoglobin levels in the hand's vascular system.
- **Thermography:** Thermal cameras detect the heat emitted by blood vessels, providing another method to visualize and analyze vascular patterns.

2) Image processing and analysis

- **Preprocessing:** Involves enhancing the captured images, removing noise, and improving contrast to highlight the vascular structures.
- **Feature Extraction:** Key features such as the width, length, and branching patterns of veins are extracted. These features can be used for biometric identification or medical diagnosis.
- **Pattern Recognition:** Advanced algorithms, often powered by machine learning, are used to recognize and classify vascular patterns. For example, CNNs can be trained to detect anomalies in the vascular geometry that might indicate disease.

3) 3D Reconstruction (Advanced):

- **3D Modeling:** For a more comprehensive analysis, 3D models of the vascular network can be constructed using multiple 2D images taken from different angles. This approach is useful in surgical planning or in-depth diagnostic procedures.

Challenges and Considerations

- **Image Quality:** The accuracy of vascular geometry analysis heavily depends on the quality of the imaging system and the clarity of the images captured.
- **Variability:** Factors such as age, temperature, and physical activity can affect the appearance of vascular patterns. Algorithms need to account for these variations.
- **Ethical Concerns:** In biometric applications, there are privacy and security concerns related to the storage and use of vascular pattern data.
- **Regulatory Approval:** In medical diagnostics, the technology must be validated and approved by relevant regulatory bodies to ensure safety and efficacy.

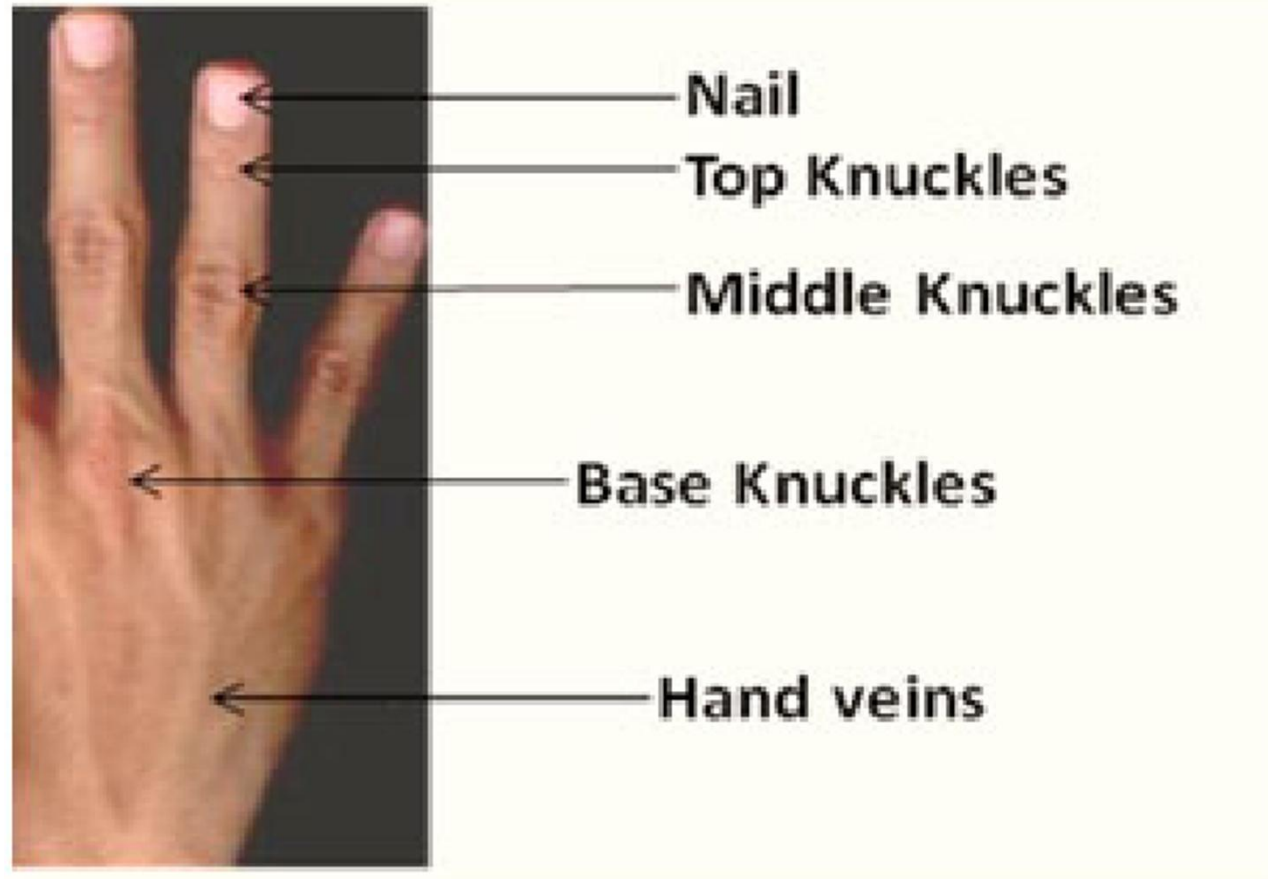
Potential Future Developments

- **Integration with Wearable Devices:** Future applications could involve wearable devices that continuously monitor vascular health, providing real-time data and alerts.
- **AI-Driven Diagnostics:** Further advancements in AI could lead to more accurate and automated diagnosis of vascular-related diseases using hand vascular geometry.
- **Personalized Medicine:** By analyzing an individual's vascular geometry, treatments could be tailored to their specific circulatory system, enhancing the effectiveness of interventions.

Knuckle biometric

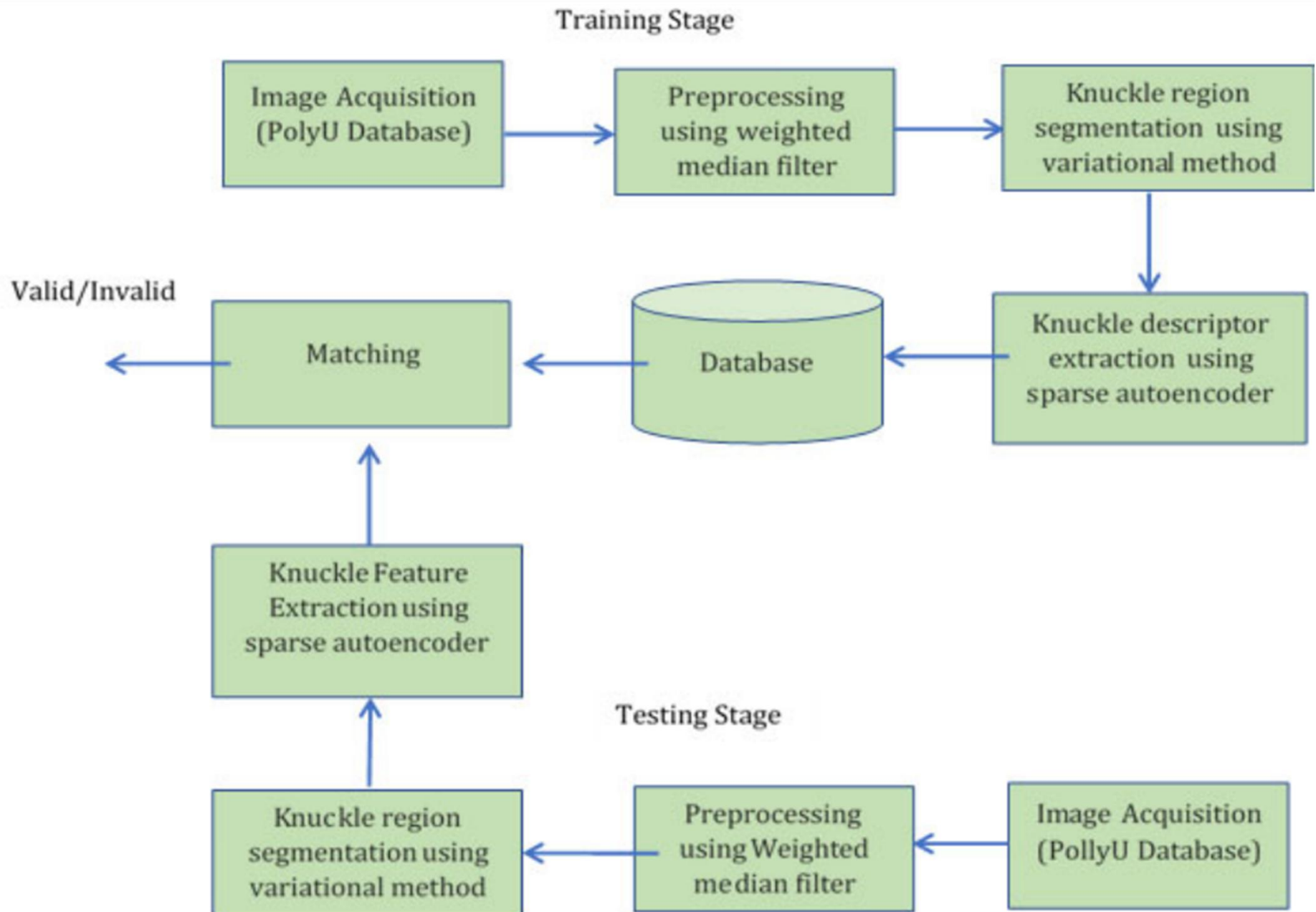
- Finger knuckle surface is one of the emerging hand-based biometric traits for personal identification.
- These unique patterns of finger knuckle surface have greater potentiality toward the distinctive identification of individuals

Sample Finger Knuckle Biometric Representation



The device used to collect finger
knuckle print images





- The first step of the work is to eliminate the noise from the image.
- The collected finger knuckle images having different type of noises due to the capturing process, name, image shake and so on.
- The inconsistent pixel details are reducing the authentications system and reduces the entire system security. So, the noise present in the finger knuckle image must be removed by applying the weighted median filter

- Initially, the pixels are checked against the threshold value.
- The threshold value is determined by analyzing the pixel range from the image.
- If the pixel value is deviated from the threshold value, it should be replaced by the weighted median value.
- During this process, the weighted value is allocated depending on the user choice. When the highest value is chosen, the image has been smoothened effectively

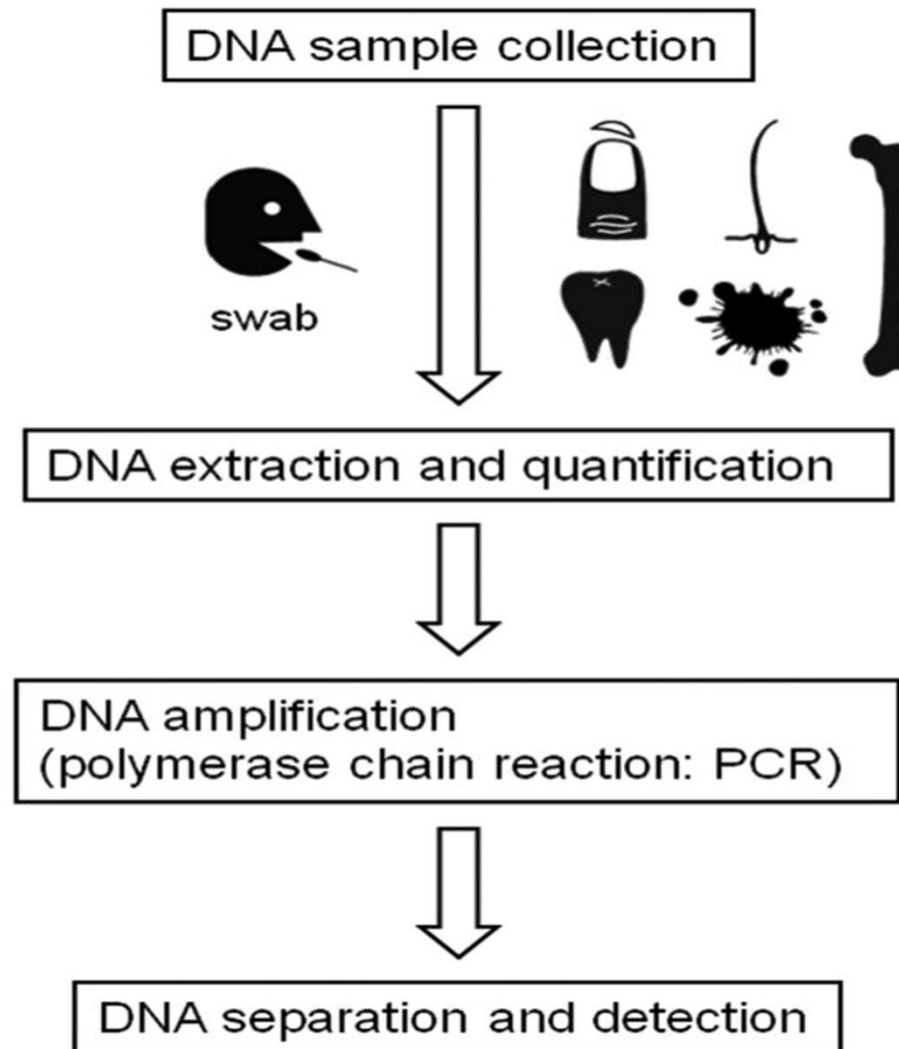
- The second step of the work is to extract the knuckle region which is done by using the variational approach.
- It selects the optimal pixels while performing grouping process also consume minimum deviation and resolve the non-convexity problem

DNA Biometrics

- deoxyribonucleic acid (DNA) provides the most reliable personal identification. It is intrinsically digital, and does not change during a person's life or after his/her death
- DNA fingerprinting is used in a variety of situations, such as criminal investigations, other forensic purposes and paternity testing.
- In these situations, one aims to “match” two DNA fingerprints with one another, such as a DNA sample from a known person and one from an unknown person.

The flow of DNA polymorphism analysis.

Refer. <https://www.intechopen.com/chapters/16506>



Dental

- Dental biometrics utilizes dental radiographs for human identification.
- The dental radiographs provide information about teeth, including tooth contours, relative positions of neighboring teeth, and shapes of the dental work (e.g., crowns, fillings, and bridges).

two main stages:

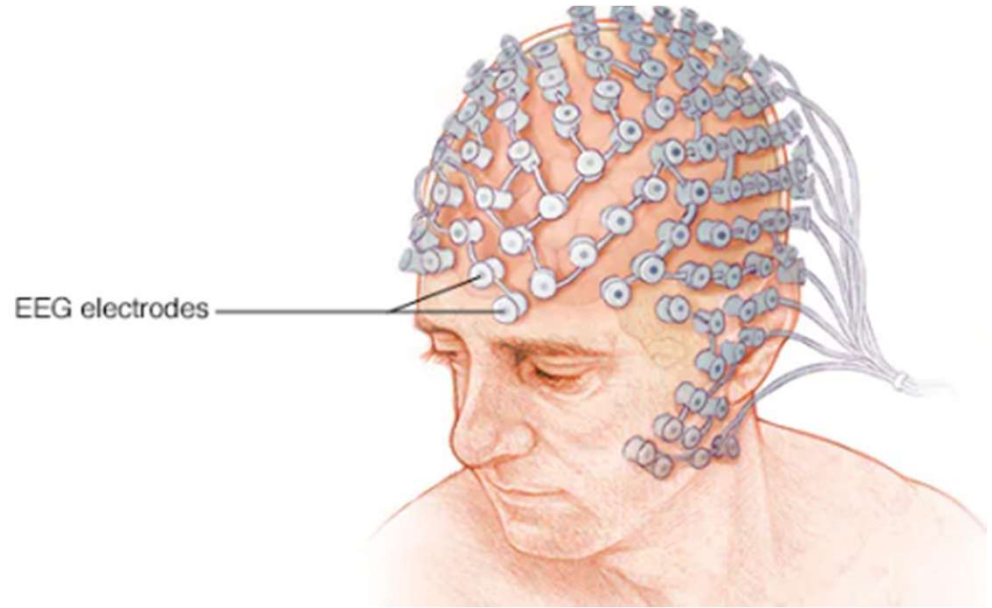
- feature extraction and matching.
 - The feature extraction stage uses anisotropic diffusion to enhance the images and a Mixture of Gaussians model to segment the dental work.
 - The matching stage has three sequential steps:
 - tooth-level matching, computation of image distances, and subject identification.
 - In the tooth-level matching step, tooth contours are matched using a shape registration method, and the dental work is matched on overlapping areas.
 - The distance between the tooth contours and the distance between the dental work are then combined using posterior probabilities.

- In the second step, the tooth correspondences between the given query (postmortem) radiograph and the database (antemortem) radiograph are established.
- A distance based on the corresponding teeth is then used to measure the similarity between the two radiographs.
- Finally, all the distances between the given postmortem radiographs and the antemortem radiographs that provide candidate identities are combined to establish the identity of the subject associated with the postmortem radiographs.

Cognitive biometrics

- Cognitive biometrics refer to methods and techniques for the authentication of a person based on the measurement of signals generated directly or indirectly from their thought processes, which reflect the mental and emotional states.

- **Cognitive biometrics** acquires information from users through collection of one or more biosignals.
- The process utilises biological based signals such as EEG or ECG as inputs to an authentication system.
- The EEG is a signal that is generated by the collective activity of neurons.
- That is, brain activity produces an electrical signal that can be recorded by placing voltage sensitive electrodes on the surface of the scalp.



With all necessary conditions, a continuous signal is recorded from each electrode positioned over the entire surface of the scalp.

A tremendous amount of data is generated during an EEG recording.

- Typically, a 10–20 electrode placement system can be used to find the location of electrodes on scalp.
- For example, it can be considered as each electrode providing a time series sampled at 0.5–1.0 KHz. generating hundreds of megabytes of data which is further analysed in order to extract useful information.

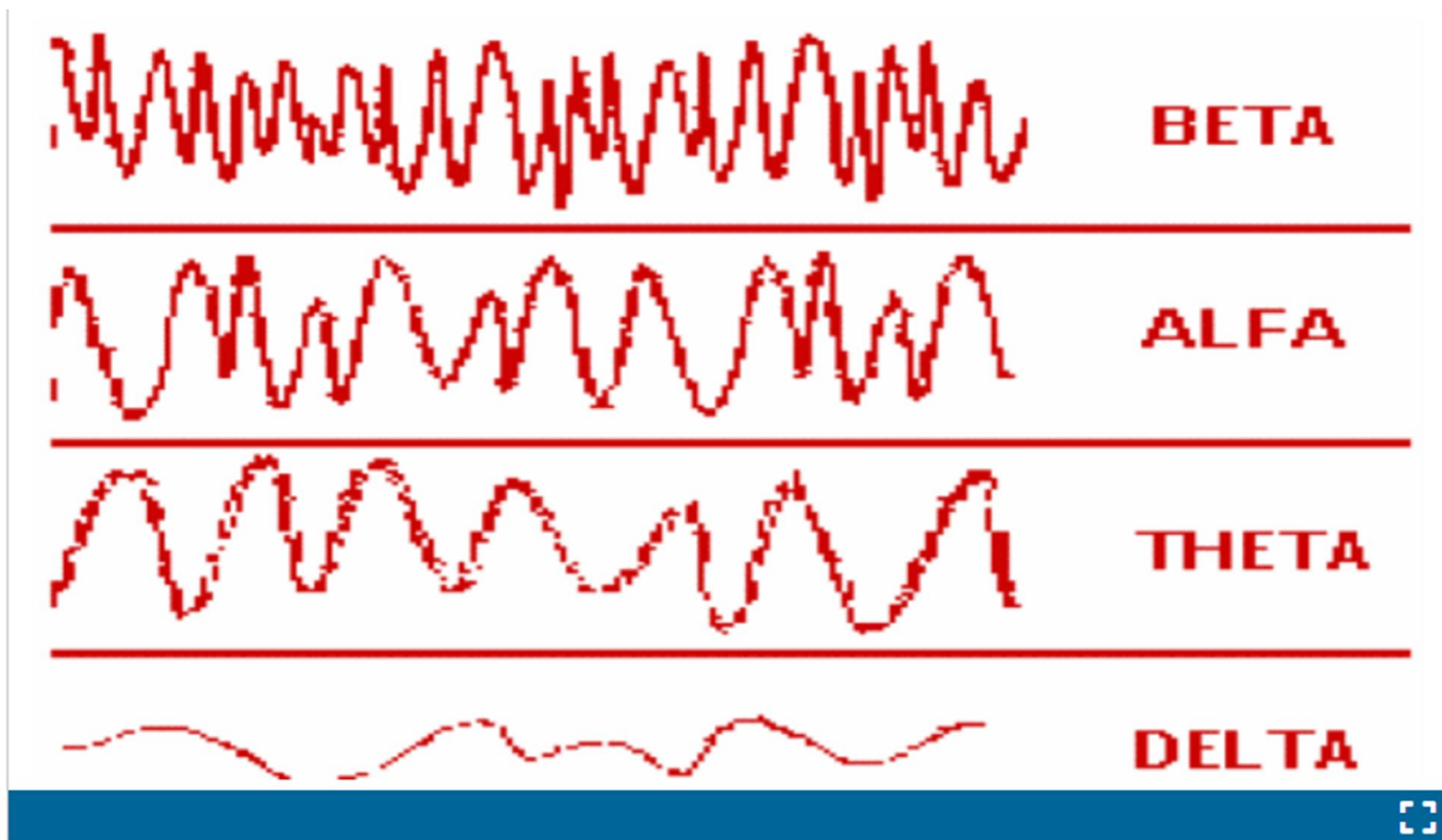


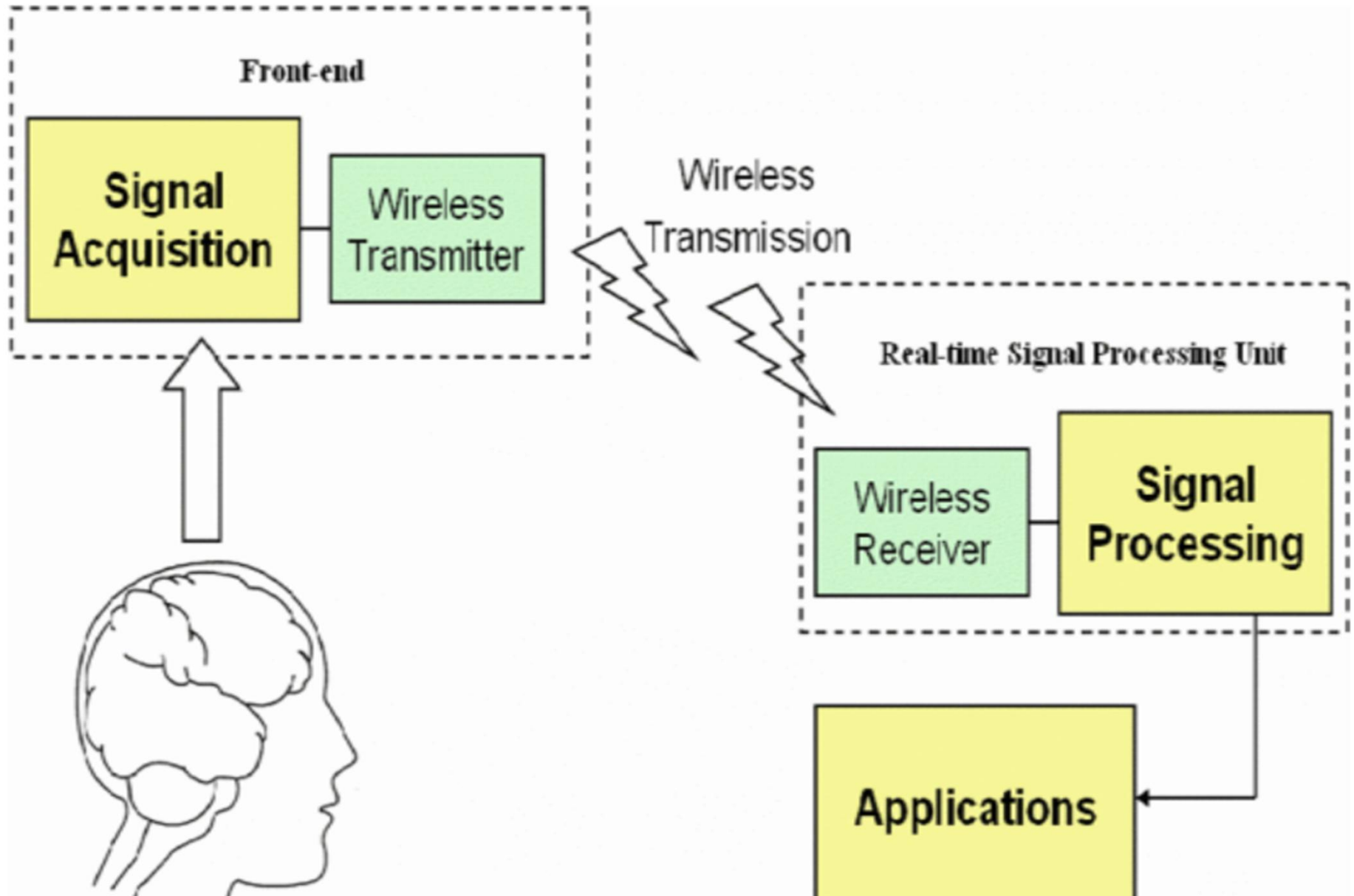
Figure 1.

Brain wave samples with dominant frequencies belonging to beta, alpha, theta, and delta band.

electroencephalogram (EEG)

- The EEG recording system consists of electrodes with conductive media, amplifiers with filters, A/D converter and a recording device.
- Electrodes pickup brain signals from the head surface, amplifiers amplify the microvolt signals into the range where they can be digitalized accurately, A/D converter converts signals from analog to digital form, and a computer can be used to store and displays obtained data.

EEG based biometric



- electrocardiogram (ECG) signals as a biometric recognition trait due to their unique properties:
- (1) liveness detection: Since the ECG is a recording of the electrical activity of the heart, ECG signals can only be acquired from living individuals

- 2) high security: ECG signals are extremely difficult to counterfeit and consequently, a technology to artificially produce them has not been developed yet
- (3) combined information: the analysis of ECG signals can give us information regarding the identity of a person, as well as heart conditions and emotional and physical status